Cognitive Neurodynamics (2025) 19:1
https://doi.org/10.1007/s11571-024-10205-x

RESEARCH ARTICLE

Neural oscillations predict flow experience

Bingxin Lin[1] · Baoshun Guo[2] · Lingyun Zhuang[1] · Dan Zhang[1] · Fei Wang[1,3]

Received: 2 August 2024 / Revised: 1 October 2024 / Accepted: 10 October 2024 / Published online: 31 December 2024
© The Author(s), under exclusive licence to Springer Nature B.V. 2024

Abstract
Flow experience, characterized by immersion in the activity at hand, provides a motivational boost and promotes positive behaviors. However, the oscillatory representations of flow experience are still poorly understood. In this study, the difficulty of the video game was adjusted to manipulate the individual's personalized flow or non-flow state, and EEG data was recorded throughout. Our results show that, compared to non-flow tasks, flow tasks exhibit higher theta power, moderate alpha power, and lower beta power, providing evidence for a focused yet effortless brain pattern during flow. Additionally, we employed Lasso regression to predict individual subjective flow scores based on neural data, achieving a correlation coefficient of 0.571 ($p < 0.01$) between the EEG-predicted scores and the actual self-reported scores. Our findings offer new insights into the oscillatory representation of flow and emphasize that flow, as a measure of individual experience quality, can be objectively and quantitatively predicted through neural oscillations.

Keywords Flow experience · Video game · Oscillatory representation · Machine learning · Quantitative prediction

Introduction

When someone is deeply immersed in the activity at hand, they may experience flow, a unique mental state characterized by phenomenological elements such as effortless control, altered time perception, and self-transcendence (Csikszentmihalyi 1990). Flow has been observed in a variety of activities, including reading (Thissen et al. 2022), sports (Wollseiffen et al. 2016), music (Loepthien and Leipold 2022), learning (Yazidi et al. 2020), and video gaming (Klasen et al. 2012; Zhang and Wang 2024). This state is optimal and beneficial, as individuals who frequently experience it in daily life are more resilient to stress (Mao et al. 2020) and enjoy higher levels of well-being (Csikszentmihalyi 1999). Furthermore, it provides a motivational basis for engaging in certain activities (Rheinberg 2020) and can be leveraged to promote positive behaviors.

Given the unique nature of flow as a state of consciousness, an intriguing question is what brain mechanisms are responsible for it. To address this question, neuroscience research on flow has witnessed a boom over the past decade (for a recent review, see Alameda et al. 2022). A milestone study by Ulrich et al. (2014) utilized fMRI to investigate the neural correlates of flow during a mental arithmetic task. This study was based on the premise that a flow state is likely to arise when the challenge of a task is balanced with an individual's skill level. Accordingly, the experiment adjusted the task difficulty to match participants' skill levels by altering the complexity of numerical operations—either adding or removing one- or two-digit numbers—to trigger flow experiences. The results indicated that flow was associated with increased activation in the inferior frontal gyrus and putamen, and decreased activation in the amygdala and medial prefrontal cortex. Building on this work, Katahira et al. (2018) examined the Electroencephalography (EEG) correlates of the flow state as participants completed mental arithmetic tasks in three conditions (i.e., Boredom, Flow, and Overload). They found increased frontal theta rhythm in the Flow and Overload conditions compared to the Boredom condition, suggesting greater cognitive control was

Bingxin Lin and Baoshun Guo contributed equally to this work.

✉ Fei Wang
wf3126@mail.tsinghua.edu.cn

[1] Department of Psychological and Cognitive Sciences, Tsinghua University, Beijing 100084, China

[2] School of Marxism, Tsinghua University, Beijing 100084, China

[3] Positive Psychology Research Center, Tsinghua University, Beijing 100084, China

required during the Flow and Overload conditions. On the other hand, there was an increase in moderate frontocentral alpha rhythm in the Flow condition compared to the Boredom condition (i.e., low difficulty). They attributed this to the task's difficulty level, as the results showed higher alpha power in the Overload condition (i.e., high difficulty) than in the Flow condition (i.e., moderate difficulty). Subsequent researchers using similar arithmetic tasks have supported the idea that alpha and theta rhythms are potential neural correlates of the flow state (Knierim et al. 2018, 2021).

In contrast to arithmetic tasks, video games are particularly well-suited for triggering flow experience due to their high levels of immersive enjoyment (Michailidis et al. 2018). Studies combining video games with fMRI have revealed that the flow state correlates with increased activity in specific brain structures, such as those involved in cognitive control, the reward system, sensorimotor networks, and the default mode network (Klasen et al. 2012; Huskey et al. 2018a, b; Ju and Wallraven 2019). However, fMRI tasks can make it more difficult for participants to experience flow due to the MRI scanner's noisy and claustrophobic environment. In contrast, EEG tasks in a laboratory setting more closely resemble real-life situations of video gaming. Therefore, the current study employed a video game paradigm to induce the flow experience, with EEG data recorded continuously throughout, aiming to examine how EEG rhythms characterize flow experience.

It raises the interesting question of whether the nature of flow is a discrete or continuous state (Csikszentmihalyi and Csikszentmihalyi 1992; Cairns et al. 2014; Michailidis et al. 2018). Some studies investigated flow as a discrete state by comparing the condition of flow, characterized by a challenge-skill balance, to non-flow conditions, marked by a challenge-skill imbalance (Huskey et al. 2018b; Katahira et al. 2018). However, other research treated flow as a continuous variable that can be used to characterize experiential quality (de Manzano et al. 2010; Wang et al. 2023). In this study, we analyze flow experience from both perspectives: as a discrete state and as a continuous variable. First, we compare brain response patterns in flow versus non-flow conditions to identify specific neural correlates. Subsequently, we treat the intensity of flow induced by gaming as a quantitative variable and assess the effectiveness of objectively measured neural responses in predicting self-reported flow scores. This dual approach allows for a nuanced understanding of the flow experience and its underlying oscillatory mechanisms.

The current study aims to explore the brain activity associated with game flow using EEG technology and to evaluate the potential of neural indicators to predict subjective flow levels. We used Tetris, a well-known and established video game (Keller and Bless 2008; Harmat et al. 2015; Hang et al. 2024). A 64-channel EEG was recorded while participants played the Tetris game in three conditions (Flow, Easy, and Difficult). We propose the following two hypotheses. First, based on the need for a focused yet effortless brain during flow, we hypothesize an increase in theta activity (higher cognitive control; Katahira et al. 2018; Knierim et al. 2018), a moderate alpha power (optimal level of cognitive load; Katahira et al. 2018; Knierim et al. 2021), and a reduction in beta power (non-hypervigilant cognitive processing; de Carvalho et al. 2015; Wollseiffen et al. 2016) under flow conditions compared to non-flow conditions. Second, we propose that objectively measured neural oscillations can quantitatively predict the degree of subjective flow experience in individuals.

Methods

Ethical statement

The study protocol was conducted in accordance with the Declaration of Helsinki and received approval from the local Ethics Committee of Tsinghua University. All participants provided written informed consent before the experiment.

Participants

Twenty-eight, right-handed college students were recruited for the study. Two participants were excluded from the analysis due to the large EEG artefacts. The twenty-six remaining participants (5 female, age: mean ± SD = 23.35 ± 3.17) reported having normal or corrected-to-normal vision and no psychiatric/neurological diseases. Upon completion of the experiment, participants received CNY ¥120 (approximately USD $16.00).

Procedure

This study employed the Tetris game to induce Flow. The initial version of the Tetris game program was obtained from the website https://github.com/k0626089/PYTRIS, which is licensed under the MIT license. In accordance with the parameters permitted by the program, a secondary development was conducted on the experimental procedure to meet the requirements of this study.

The experiment consisted of an adaptation phase and a formal experimental phase (Fig. 1). During the adaptation phase, Tetris started at a default speed, which was then adjusted during the game to match the participant's performance, evaluated every 30th piece (as in Keller and Bless 2008). Specifically, if the participant succeeded in creating five or more complete rows using the previous 30 pieces, the

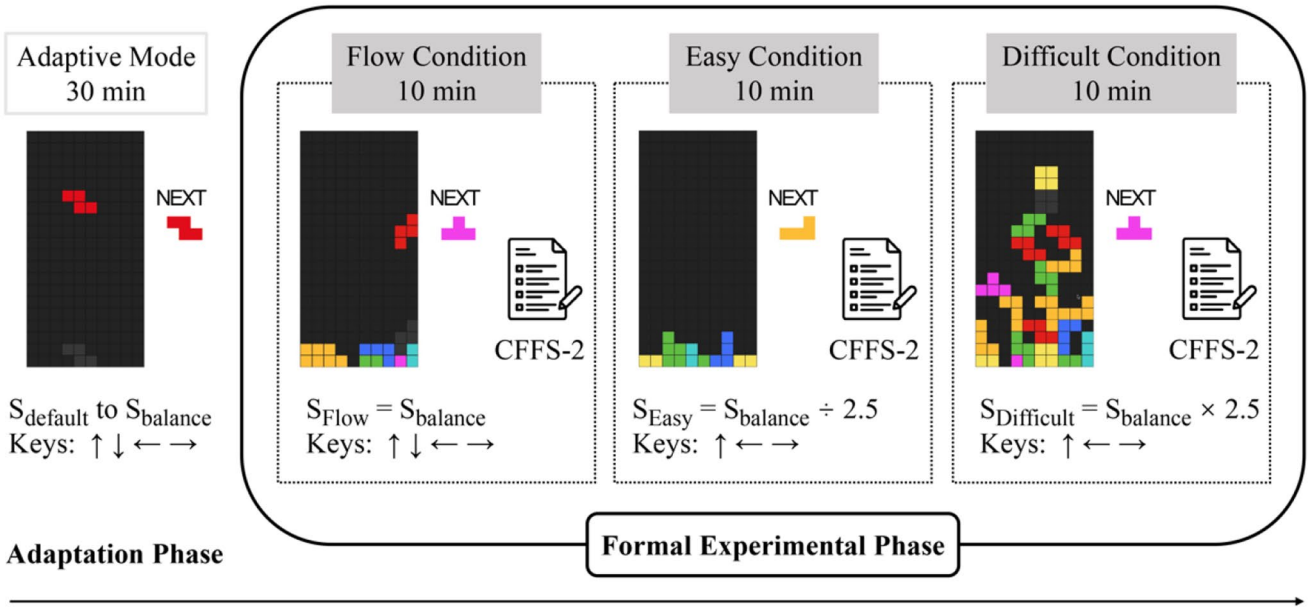


Fig. 1 Experimental Procedure. First is the adaptation phase, where Tetris starts at a default speed, adjusted based on the participant's performance. Participants could instantly drop the falling brick using the down arrow key. After 30 min, each participant received a speed tailored to their skill level ($S_{balance}$). Following this, the formal experimental phase consisted of 3 conditions in a random order. In the Flow condition, the initial speed was set to $S_{balance}$, and other game settings were consistent with the adaptation phase. The initial speed of the Easy condition was set to 0.4 times the $S_{balance}$, and that of the Difficult condition was also set to 2.5 times the $S_{balance}$. It is important to note that the speeds in the Easy and Difficult conditions remained constant throughout, and participants could not use the down arrow key. After completing each task of the three conditions, participants were required to fill out the CFFS-2 separately to report their subjective experience of flow during the preceding task

speed of the falling bricks was increased by a fixed value; if the participant managed to create three or fewer complete rows using the previous 30 pieces, the speed was decreased by the corresponding fixed value. Additionally, participants could use the down arrow key to instantly drop the falling brick, allowing for further control and challenge-skill adjustment. Finally, a $S_{balance}$ (the optimal speed) for each participant was determined after 30 min of adaptive mode.

Following the adaptation phase, the formal experimental phase commenced. This phase consisted of tasks under three conditions: Flow, Easy, and Difficult, presented in random order. Each condition lasted 10 min. In the Flow condition, the initial speed was set to $S_{balance}$, and the speed was adjusted based on the participant's real-time performance. Additionally, participants could use the down arrow key to instantly drop the falling brick, which was consistent with the adaptation phase. In the Easy condition, the speed was set to 0.4 times the $S_{balance}$; in the Difficult condition, the speed was set to 2.5 times the $S_{balance}$. In both the Easy and Difficult conditions, the speed remained constant throughout regardless of the participant's performance, and the down arrow key could not be used to drop the falling brick instantly.

After each condition, participants were required to fill out the Chinese version of the Flow State Scale-2 (CFSS-2) (Liu et al., 2012) to self-report their flow experience during the preceding condition. The CFSS-2 comprises 36 items rated on a 5-point scale. Total scores represent the level of flow experience for that condition, with higher scores indicating a greater level of flow experience.

EEG recording and preprocessing

The EEG signals were recorded from 64 channels with an amplifier of Brain Product (Brain Products GmbH, Steingrabenstr, Germany) at a sampling rate of 500 Hz. The electrodes were positioned according to the extended international 10–20 system with a reference electrode at TP9. Electrode impedances were maintained below 10 kΩ throughout the experiment.

All preprocessing of EEG data was performed offline using MATLAB and the EEGLAB toolbox. To avoid potential onset and offset effects, the first and last 30 s of each participant's 10-minute EEG signal were excluded from further analysis, resulting in a total duration of 540 s. Next, EEG data were re-referenced using the average of all channels. EEG data underwent notch filtering to remove the 50 Hz powerline interference, followed by band-pass filtering between 0.1 Hz and 45 Hz. Independent components analysis was performed to remove eye-movement-related activity and motor artefacts according to visual inspection. Afterward, the EEG data were segmented into 2-second

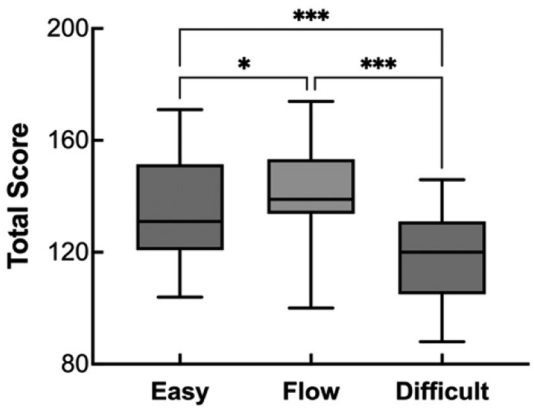


Fig. 2 Differences in self-reported flow scores between three conditions. Error bars denote the standard error. *: $p < 0.05$, ***: $p < 0.001$. p-values were corrected by FDR

epochs, and spectral analysis was conducted using fast Fourier transform to obtain the power at each frequency. These EEG powers were further transformed into relative powers by dividing by the sum of the absolute powers across all frequencies (i.e., 1–45 Hz). Finally, the relative EEG powers were averaged over the delta (1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–45 Hz) bands, respectively, resulting in relative EEG powers for these five frequency bands.

Results

Behavioral data

A repeated-measures ANOVA was conducted on the total scores of the flow questionnaire reported by the participants (Fig. 2). There were significant differences among the three conditions ($F(2, 50) = 26.858$, $p < 0.001$, $\eta_p^2 = 0.518$). Post hoc comparisons revealed that the self-reported scores for the Flow condition ($M = 141.308$, $SD = 15.281$) were significantly higher than those for the Easy condition ($M = 135.231$, $SD = 17.842$; $p = 0.037$, FDR corrected), and the Difficult condition ($M = 120.000$, $SD = 15.723$; $p < 0.001$, FDR corrected). Additionally, the scores for the Easy condition were significantly higher than those for the Difficult condition ($p < 0.001$, FDR corrected).

EEG data

As shown in Figs. 3 and 4, the EEG power spectrum and topographic maps indicate that brain responses appear to differ under various conditions. Analyses to compare the differences among conditions were performed separately for each frequency band using repeated measures ANOVA, considering three conditions (Flow, Easy, Difficult).

For the delta band, no significant main effect was found ($F(2, 50) = 1.347$, $p = 0.269$, $\eta_p^2 = 0.051$), indicating no differences between any pairwise comparisons among the Flow ($M = 0.0311$, $SD = 0.0035$), Easy ($M = 0.0307$, $SD = 0.0036$), and Difficult conditions ($M = 0.0314$, $SD = 0.0038$).

For the theta band, a significant main effect was observed ($F(2, 50) = 22.822$, $p < 0.001$, $\eta_p^2 = 0.477$). Post hoc comparisons indicated that an increase of theta power in the Flow condition ($M = 0.0204$, $SD = 0.0034$) compared to both the Easy ($M = 0.0183$, $SD = 0.0028$; $p < 0.001$, FDR corrected) and Difficult conditions ($M = 0.0195$, $SD = 0.0027$; $p < 0.001$, FDR corrected). Furthermore, theta power of the Difficult condition was significantly greater than the Easy condition ($p = 0.007$, FDR corrected).

For the alpha band, a significant main effect was found (Greenhouse–Geisser's correction: $F(1.617, 40.425) = 7.077$, $p = 0.004$, $\eta_p^2 = 0.221$). Post hoc comparisons revealed that the alpha power of the Easy condition ($M = 0.0159$, $SD = 0.0027$) was significantly higher than the Difficult condition ($M = 0.0150$, $SD = 0.0019$; $p = 0.013$, FDR corrected). While the Flow condition ($M = 0.0154$,

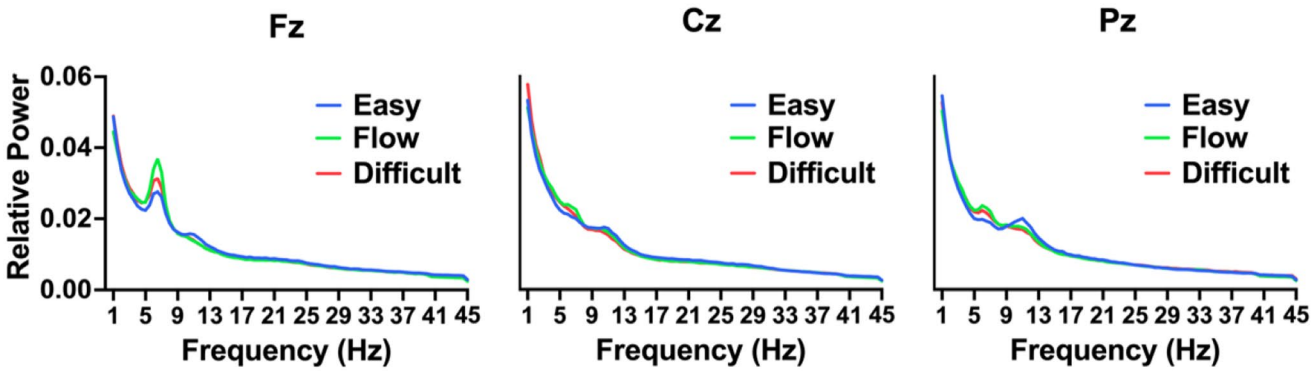


Fig. 3 EEG Power Spectrum. The distribution of relative EEG power across the 1–45 Hz frequency range is shown for Easy, Flow, and Difficult conditions, with Fz, Cz, and Pz as examples

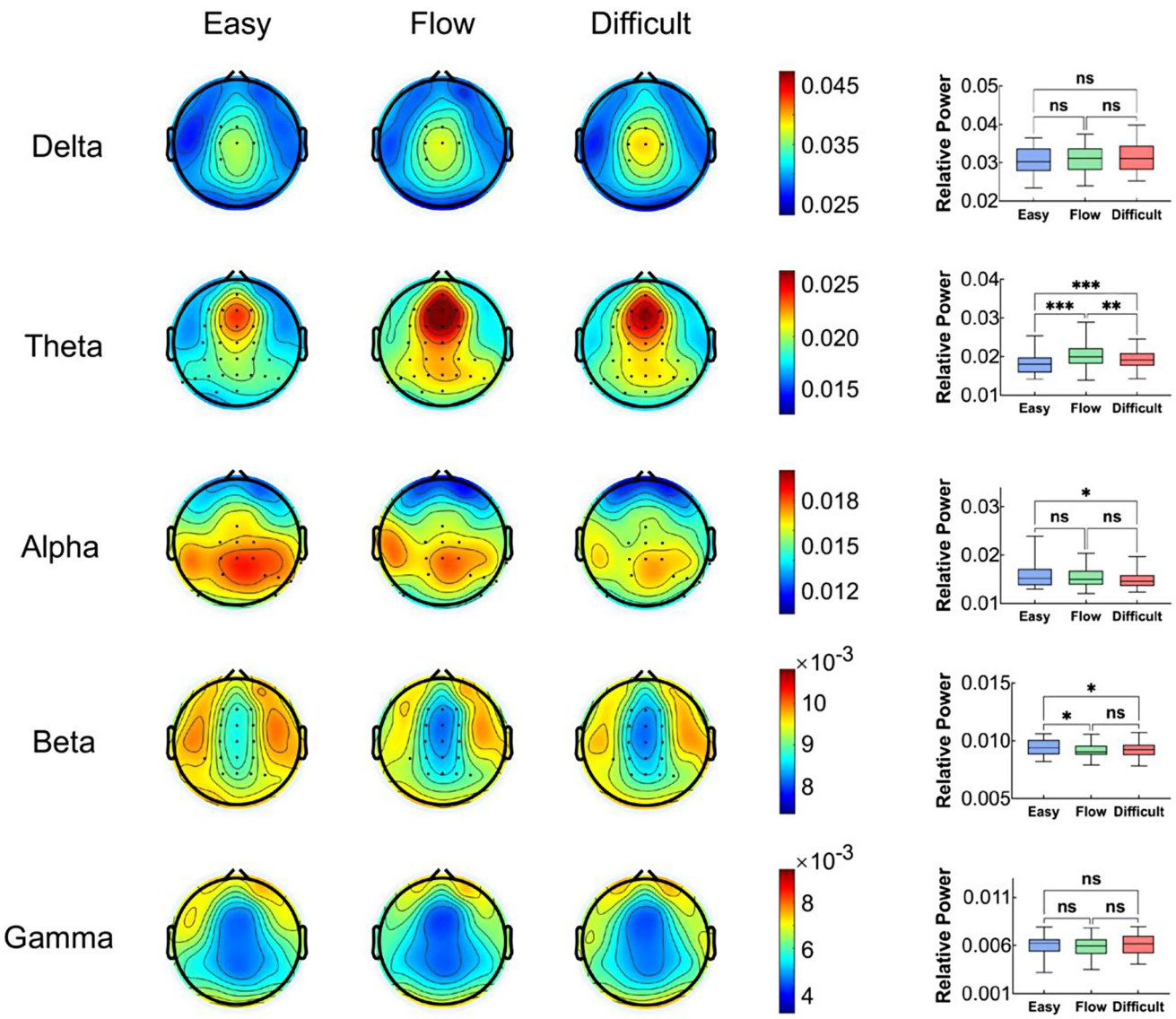


Fig. 4 Topographic maps of five bands under three conditions, and comparison of the differences between different conditions for each frequency band. The channels presented in the topographic maps have been corrected by FDR ($p < 0.001$). Error bars denote the standard error. ns: $p > 0.05$, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$. p-values were corrected by FDR

$SD = 0.0022$) did not significantly differ from the other two conditions ($ps > 0.05$).

For the beta band, a significant main effect was observed ($F(2, 50) = 5.991$, $p = 0.005$, $\eta_p^2 = 0.193$). Post hoc comparisons indicated that an increase of beta power in the Easy condition ($M = 0.0094$, $SD = 0.0006$) compared to both the Flow condition ($M = 0.0092$, $SD = 0.0006$; $p = 0.018$, FDR corrected) and Difficult conditions ($M = 0.0092$, $SD = 0.0007$; $p = 0.026$, FDR corrected). While there was no significant difference between the Flow condition and the Difficult condition ($p = 0.619$).

For the gamma band, no significant main effect was found ($F(2, 50) = 1.491$, $p = 0.235$, $\eta_p^2 = 0.056$), indicating no differences between any pairwise comparisons among the Flow ($M = 0.0059$, $SD = 0.0011$), Easy ($M = 0.0061$, $SD = 0.0011$), and Difficult conditions ($M = 0.0061$, $SD = 0.0011$).

Correlations between band powers and subjective flow scores

Pearson's correlations were calculated for the relationships between relative band powers and subjective flow scores under three conditions (Table 1; Fig. 5). The results revealed significant positive and negative correlations between flow scores and theta power ($r = 0.392$, $p = 0.048$) and gamma power ($r = -0.405$, $p = 0.040$), respectively, under the Flow condition.

Table 1 Pearson's correlations between band powers and subjective flow scores

	Easy condition		Flow condition		Difficult condition	
	r	*p*	*r*	*p*	*r*	*p*
Delta	0.249	0.219	0.388	0.050	0.132	0.522
Theta	0.239	0.239	**0.392**	0.048	-0.160	0.434
Alpha	-0.086	0.675	0.118	0.567	-0.112	0.585
Beta	-0.114	0.580	-0.374	0.060	0.097	0.637
Gamma	-0.206	0.312	**-0.405**	0.040	-0.011	0.957

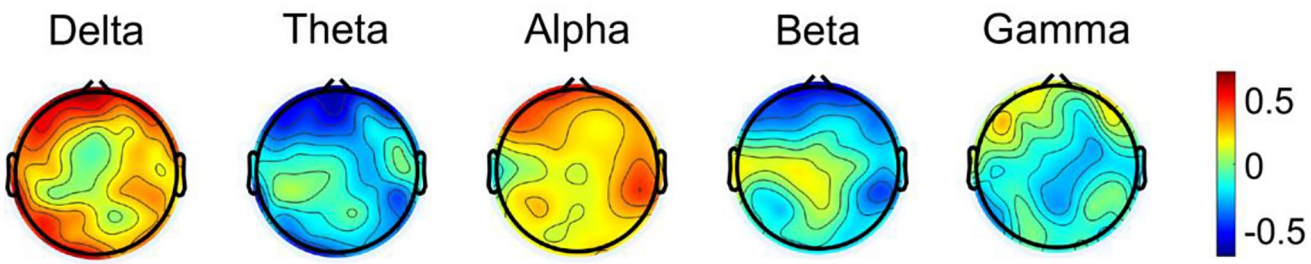


Fig. 5 Topographic maps of correlations in the Flow condition. Pearson's correlations between delta, theta, alpha, and beta band powers and self-reported flow scores are shown

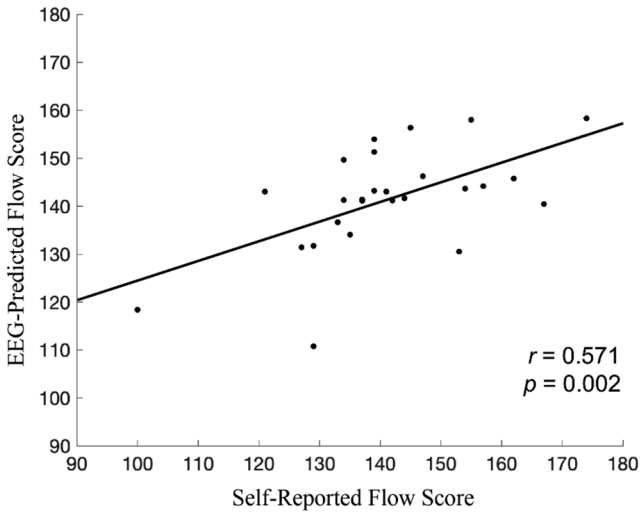


Fig. 6 Scatter plot of EEG-predicted and self-reported flow scores in the Flow condition. The horizontal axis represents flow scores self-reported by the participants, while the vertical axis represents the flow scores predicted using EEG data through Lasso regression. Each point represents the self-reported and predicted flow scores for each participant in the Flow condition

Quantitative prediction of subjective flow scores from EEG indicators

In the previous analyses, we identified distinct brain response patterns across different conditions. Next, we aim to quantitatively predict subjective flow scores using regression analysis. As shown in Fig. 2, the Flow condition successfully induced the flow experience to a considerable extent, so the following analysis only includes data of Flow condition. To reduce data dimensionality while avoiding overfitting, we employ the Lasso regression method. The relative power of five frequency bands are used as features, and the leave-one-out method is utilized for training. As shown in Fig. 6, the correlation coefficient between the predicted and actual flow scores of the participants is $r = 0.571$ ($p = 0.002$), showing that it is possible to effectively and quantitatively predict flow scores using EEG. The coefficients for the five frequency bands (delta, theta, alpha, beta, gamma) are 10.29, 0, 0, -11.28, and −30.26, respectively, which roughly align with the results of the correlation analysis. Specifically, the delta band has a positive effect on the prediction of self-report scores, while the beta and gamma bands have a negative effect. However, the theta band did not show any contribution. This may be due to the different aims of correlational analysis and the regression model: the former examines the relationship between a single frequency band and the target variable, whereas the latter considers multiple features simultaneously, meaning that the contribution of a particular band may not be significant in a multivariate model.

Discussion

The current study employed the Tetris game to trigger flow experience and explored the brain oscillations underlying it. We found that varying levels of game difficulty induced different intensities of flow experience, each associated with distinct patterns of neural oscillations. Specifically, intense flow states correlate with increased activity in the theta band, reflecting heightened cognitive control; moderate activity in the alpha band, suggesting a balanced working

memory load; and decreased activity in the beta band, pointing shifts to more automatic processing. These results suggest that during flow, the brain operates in a state of focused, yet effortless engagement. Furthermore, we were able to use these oscillation patterns to quantitatively predict the self-reported score of flow.

We observed that theta power (especially frontal midline theta, Fig. 3) was highest under the Flow condition, followed by the Difficult condition, and lowest under the Easy condition, forming an inverted U-shape pattern between difficulty and flow. Enhanced activity in the frontal theta has been frequently reported to be related to increased attention, cognitive control, and working memory capabilities (Jensen and Tesche 2002; Sauseng et al. 2005; Lagopoulos et al. 2009; Hsieh and Ranganath 2014; Clayton et al. 2015; Cona et al. 2020; Zhang et al. 2022; Ursino et al. 2023; Liu et al. 2024), the present theta results suggest that flow task indeed demands more sustained attention and cognitive control compared to easy task, similar to observations in arithmetic tasks that trigger flow (Katahira et al. 2018; Knierim et al. 2018). However, we observed a reduction of theta in the difficult condition that was excessively challenging. This contrasts with studies on arithmetic tasks, where theta wave activity remained stable across both flow and difficult/overload, which also necessitated significant cognitive control and attention. This may be due to the different nature of the arithmetic and video gaming tasks. Because the inverted U-shaped relationship was also observed in other Tetris studies (Ewing et al. 2016), it has been proposed that frontal theta could serve as an indicator for detecting the 'tipping point' at which effort is withdrawn due to cognitive overload. When at optimal difficulty, individuals maximize their cognitive control abilities, their attention is highly focused, and they experience a deep sense of immersion, characterized by the highest theta power.

Furthermore, we found that the flow condition was linked to moderate alpha activation, although there was no statistically significant difference compared to easy or difficult conditions. Notably, alpha activation was higher in the easy condition than the difficult condition, suggesting a suppression of alpha power as task difficulty increased. Previous research has reported inconsistent results regarding alpha. For instance, arithmetic tasks (Katahira et al. 2018) have demonstrated an increase in alpha power moving from a state of boredom to flow, and then to overload, attributed to the varying levels of task difficulty. Additionally, a study using a dual-task paradigm (Castellar et al. 2019) —a video game combined with an auditory oddball task—revealed higher alpha power in the flow state than in the boredom or overload states, resulting in an inverted U-shape, and an increase in response times for the secondary task, indicating inhibition of processing irrelevant information. This points towards alpha's role in inhibitory top-down cognitive control processes. There are also findings consistent with our study that reported moderate alpha activation during tasks like arithmetic (Knierim et al. 2021) and Tetris (Ewing et al. 2016). A plausible explanation for these phenomena is that alpha power is suppressed as task difficulty ranges from low to high to excessively high. Being associated with task-specific cognitive processes, the variations in the alpha band might reflect the objective level of cognitive demand imposed by current tasks (Pfürtscheller 1992; Klimesch 1999; Ewing et al. 2016). High alpha power indicates decreased neural activity, suggesting a 'deficient demand' for a given task due to lower cognitive demand. Conversely, low alpha power suggests a state of 'excessive demand,' where difficult tasks require substantial cognitive resources. Thus, the moderate level of alpha activation seen in the Flow condition likely indicates that these tasks strike a balance in cognitive demand—neither too easy nor too difficult—allowing individuals to engage with them effectively.

There is also a notable finding that beta activity is more pronounced during the Easy condition than during Flow or Difficult conditions. Intriguingly, in previous research, lower beta activity, which correlates with heightened flow experiences (Léger et al. 2014), tends to increase as task difficulty rises (Klarkowski 2017). This appears to contradict our results. However, as a marker of hypervigilance, an increase in beta activity signals a higher level of vigilance in a task (Freeman et al. 1999; de Carvalho et al. 2015), which is also linked to personal distress (Woodruff et al. 2011). Previous research indicates that as individuals become more relaxed and less vigilant during a task, alpha power increases while beta power decreases (Davidson et al. 1990; Freeman et al. 1999; Prinzel et al. 2000; Berka et al. 2007), and they report being more cognitively absorbed (Léger et al. 2014). Consistent with these findings, we observed that the Flow condition displayed moderately high alpha power and reduced beta power. This suggests that individuals in flow are likely in an internally motivated state, engaging in relaxed, non-hypervigilant cognitive processing.

In summary, when treating flow as a discrete state, this study reveals clear differences in brain activity between states of flow and non-flow. During the flow state, increased theta activity suggests enhanced cognitive control. Simultaneously, moderate alpha activity reflects a balanced cognitive load, while decreased beta activity indicates relaxed, non-hypervigilant cognitive processing. Collectively, these findings—high theta, moderate alpha, and low beta activity—serve as specific EEG indicators of the flow state, characterizing the brain as being intensely focused yet effortlessly engaged.

Despite views categorizing flow as an "all-or-nothing" experience, some scholars advocate for a continuous

approach to reflect the intensity of flow experiences (de Manzano et al. 2010; Wang et al. 2023). This study proposes using neural oscillations to predict subjective flow ratings quantitatively, offering a new method to objectively assess the subjective flow experience. First, we conducted correlational analyses between self-reported flow scores and EEG indicators and found associations between high subjective flow score and high theta power, as well as low gamma power. The theta result may be interpreted as representing cognitive control during tasks through theta activity (Klimesch 1999; Cavanagh and Frank 2014). This is consistent with findings in arithmetic tasks where a balance between challenge and skill is positively correlated with theta power (Katahira et al. 2018). Additionally, this trend aligns with findings obtained from comparisons between conditions. Interestingly, we found that lower gamma power was associated with higher subjective flow ratings, although this phenomenon was not observed in the comparison of differences between conditions. The relationship between gamma and consciousness has been extensively reported (Lehmann et al. 2001; John 2002; Panagiotaropoulos et al. 2012), and the negative correlation between flow and gamma neatly illustrates a crucial dimension of flow experience: loss of self-consciousness (Csikszentmihalyi and Nakamura 2018).

Finally, we utilized the Lasso regression method to predict self-reported flow scores based on neural oscillations, achieving a correlation coefficient of 0.571 between the EEG-predicted and actual flow scores. This demonstrates the feasibility of quantitatively predicting subjective flow ratings through neural oscillations, providing an effective method for objectively measuring the flow experiences without continuous interruptions. To achieve more accurate and real-time predictions, future research could consider combining affective computing with neural oscillation data (Ding et al. 2021; Shen et al. 2023), to more precisely predict and understand subjective flow experiences of individuals.

Although the present study offers new findings, it still has some limitations worth noting. First, due to the high temporal resolution but low spatial resolution of EEG, the ability to investigate brain regions involved in the predictive mechanisms of flow is limited. Furthermore, this inability prevents directly testing the three classic theoretical frameworks related to the neural basis of flow: (1) the Transient Hypofrontality Hypothesis (THH; Dietrich 2004), which proposes a shift from the explicit cognitive system to the implicit/automatic system, involving areas like the basal ganglia and cerebellum; (2) the Synchronization Theory of Flow (STF; Weber et al. 2009), which suggests that flow emerges from the cognitive synchronization of attentional and reward networks; and (3) the Internal Model of Flow (IMF; Gold and Ciorciari 2021), which highlights the importance of the cerebellum. While all three theories provide neural accounts for certain phenomenological elements of flow, no consensus has been reached. Future studies could employ MEG or multimodal approaches to conduct more fine-grained analyses of the spatiotemporal dynamics of flow. Second, previous studies have found that brain rhythms, such as gamma, may exhibit non-oscillatory components characterized by stochastic variations in time and frequency (Wang et al. 2021; Han et al. 2022). Distinguishing between oscillatory and non-oscillatory components across different frequency bands deserves further investigation in future studies to better understand how EEG rhythms may represent the experience of flow in the brain.

Conclusion

The current study investigates how flow states are represented in the brain. By employing a Tetris game, we identified an intense focus yet effortless brain activity pattern during flow, characterized by higher theta activity reflecting greater cognitive control, moderate alpha activity indicating a balanced cognitive load, and reduced beta activity signaling a state of relaxation and non-hypervigilance. Furthermore, we demonstrated that it is possible to quantitatively predict subjective flow ratings based on neural oscillations in individuals. Overall, the present findings not only deepen our understanding of the oscillatory mechanisms underlying flow but also suggest methods to assess the quality of flow experiences objectively without disruption.

Declarations

Conflict of interest The authors declare no competing interests.

References

Alameda C, Sanabria D, Ciria LF (2022) The brain in Flow: a systematic review on the neural basis of the Flow State. Cortex 154:348–364

Berka C, Levendowski DJ, Lumicao MN, Yau A, Davis G, Zivkovic VT, Olmstead RE, Tremoulet PD, Craven PL (2007) EEG correlates of Task Engagement and Mental Workload in Vigilance, Learning, and memory tasks. Aviat Space Environ Med 78(5):B231–B244

Cairns P, Cox A, Nordin AI (2014) Immersion in Digital games: review of Gaming Experience Research. Wiley, New York, NY

Castellar EPN, Antons J-N, Marinazzo D, Van Looy J (2019) Mapping attention during Gameplay: Assessment of behavioral and ERP markers in an auditory Oddball Task. Psychophysiology, 56(7), e13347

Cavanagh JF, Frank MJ (2014) Frontal Theta as a mechanism for Cognitive Control. Trends Cogn Sci 18(8):414–421

Clayton MS, Yeung N, Kadosh RC (2015) The roles of cortical oscillations in sustained attention. Trends Cogn Sci 19(4):188–195

Cona G, Chiossi F, Di Tomasso S, Pellegrino G, Piccione F, Bisiacchi P, Arcara G (2020) Theta and Alpha oscillations as signatures of Internal and External attention to delayed intentions: a magneto-encephalography (MEG) Study. NeuroImage 205:116295

Csikszentmihalyi M (1990) Flow: the psychology of optimal experience. Harper & Row, New York, NY

Csikszentmihalyi M (1999) If we are so Rich, why aren't we happy? Am Psychol 54(10):821–827

Csikszentmihalyi M, Csikszentmihalyi IS (1992) Optimal experience: psychological studies of Flow in consciousness. Cambridge University Press, Cambridge, England

Csikszentmihalyi M, Nakamura J (2018) Flow, altered States of consciousness, and human evolution. J Conscious Stud 25(11–12):102–114

Davidson RJ, Chapman JP, Chapman LJ, Henriques JB (1990) Asymmetrical Brain Electrical-Activity discriminates between psychometrically-matched verbal and spatial cognitive tasks. Psychophysiology 27(5):528–543

de Carvalho MR, Velasques BB, Freire RC, Cagy M, Marques JB, Teixeira S, Thomaz R, Range BP, Piedade R, Akiskal HS, Nardi AE, Ribeiro P (2015) Frontal Cortex Absolute Beta Power Measurement in panic disorder with Agoraphobia patients. J Affect Disord 184:176–181

de Manzano Ö, Theorell T, Harmat L, Ullén F (2010) Psychophysiology of Flow during piano playing. Emotion 10(3):301–311

Dietrich A (2004) Neurocognitive mechanisms underlying the experience of Flow. Conscious Cogn 13(4):746–761

Ding Y, Hu X, Xia Z, Liu Y-J, Zhang D (2021) Inter-brain EEG feature extraction and analysis for continuous implicit emotion tagging during video watching. IEEE Trans Affect Comput 12(1):92–102

Ewing KC, Fairclough SH, Gilleade K (2016) Evaluation of an adaptive game that uses EEG measures validated during the design process as inputs to a Biocybernetic Loop. Front Hum Neurosci 10:223

Freeman FG, Mikulka PJ, Prinzel LJ, Scerbo MW (1999) Evaluation of an adaptive automation system using three eeg indices with a visual Tracking Task. Biol Psychol 50(1):61–76

Gold J, Ciorciari J (2021) A neurocognitive model of Flow States and the role of Cerebellar Internal models. Behav Brain Res 407:113244

Han C, Shapley R, Xing D (2022) Gamma rhythms in the visual cortex: functions and mechanisms. Cogn Neurodyn 16(4):745–756

Hang Y, Unenbat B, Tang S, Wang F, Lin B, Zhang D (2024) Exploring the neural correlates of Flow experience with multifaceted tasks and a single-Channel Prefrontal EEG Recording. Sensors 24(6):1894

Harmat L, de Manzano Ö, Theorell T, Högman L, Håkan H, Ullén F (2015) Physiological correlates of the Flow experience during computer game playing. Int J Psychophysiol 97(1):1–7

Hsieh L-T, Ranganath C (2014) Frontal midline Theta oscillations during Working Memory maintenance and episodic encoding and Retrieval. NeuroImage 85:721–729

Huskey R, Craighead B, Miller MB, Weber R (2018a) Does intrinsic reward motivate Cognitive Control? A naturalistic-fmri study based on the synchronization theory of Flow. Cogn Affect Behav Neurosci 18(5):902–924

Huskey R, Wilcox S, Weber R (2018b) Network Neuroscience reveals distinct neuromarkers of Flow during Media Use. J Communication 68(5):872–895

Jensen O, Tesche CD (2002) Frontal Theta activity in humans increases with memory load in a Working Memory Task. Eur J Neurosci 15(8):1395–1399

John ER (2002) The neurophysics of consciousness. Brain Res Rev 39(1):1–28

Ju U, Wallraven C (2019) Manipulating and decoding subjective gaming experience during active gameplay: a Multivariate, whole-brain analysis. NeuroImage 188:1–13

Katahira K, Yamazaki Y, Yamaoka C, Ozaki H, Nakagawa S, Nagata N (2018) EEG correlates of the Flow State: a combination of increased Frontal Theta and Moderate Frontocentral Alpha Rhythm in the Mental arithmetic Task. Front Psychol 9:300

Keller J, Bless H (2008) Flow and Regulatory Compatibility: An Experimental Approach to the Flow Model of intrinsic motivation. Pers Soc Psychol Bull 34(2):196–209

Klarkowski MW (2017) The psychophysiological evaluation of the player experience (Doctoral dissertation). Queensland University of Technology, Queensland, Australia

Klasen M, Weber R, Kircher TTJ, Mathiak KA, Mathiak K (2012) Neural contributions to Flow Experience during Video Game playing. Soc Cognit Affect Neurosci 7(4):485–495

Klimesch W (1999) EEG Alpha and Theta oscillations reflect cognitive and memory performance: a review and analysis. Brain Res Rev 29(2–3):169–195

Knierim MT, Nadj M, Hariharan A, Weinhardt C (2018) Flow neurophysiology in knowledge work: electroencephalographic observations from two cognitive tasks. In Proceedings of the 5th International Conference on Physiological Computing Systems (pp. 42–53). Seville, Spain

Knierim MT, Berger C, Reali P (2021) Open-source concealed EEG Data Collection for Brain-Computer-interfaces-neural Observation through Openbci amplifiers with around-the-Ear Ceegrid Electrodes. Brain-Computer Interfaces 8(4):161–179

Lagopoulos J, Xu J, Rasmussen I, Vik A, Malhi GS, Eliassen CF, Arntsen IE, Saether JG, Hollup S, Holen A, Davanger S, Ellingsen O (2009) Increased Theta and Alpha EEG Activity during Nondirective Meditation. J Altern Complement Med 15(11):1187–1192

Léger P-M, Davis FD, Cronan TP, Perret J (2014) Neurophysiological correlates of cognitive absorption in an enactive training context. Comput Hum Behav 34:273–283

Lehmann D, Faber PL, Achermann P, Jeanmonod D, Gianotti LRR, Pizzagalli D (2001) Brain sources of EEG Gamma frequency during Volitionally Meditation-Induced, altered States of consciousness, and experience of the self. Psychiatry Research-Neuroimaging 108(2):111–121

Liu W, Liu X, Ji L, Watson II JC, Zhou C, Yao J (2012) Chinese translation of the flow state scale-2 and the dispositional flow scale-2: examination of factorial validity and reliability. International Journal of Sport Psychology 43(2):153–175

Liu Y, Yin H, Liu X, Zhang L, Wu D, Shi Y, Chen Y, Zhou X (2024) Alcohol Use Disorder and Time Perception: the mediating role of attention and Working Memory. Addict Biol, 29(2), e13367

Loepthien T, Leipold B (2022) Flow in Music performance and Music-Listening: differences in intensity, predictors, and the relationship between Flow and Subjective Well-Being. Psychol Music 50(1):111–126

Mao Y, Yang R, Bonaiuto M, Ma J, Harmat L (2020) Can Flow alleviate anxiety? The roles of Academic Self-Efficacy and Self-Esteem in Building Psychological sustainability and resilience. Sustainability 12(7):2987

Michailidis L, Balaguer-Ballester E, He X (2018) Flow and Immersion in Video games: the Aftermath of a conceptual challenge. Front Psychol 9:1682

Panagiotaropoulos TI, Deco G, Kapoor V, Logothetis NK (2012) Neuronal discharges and Gamma oscillations explicitly reflect visual consciousness in the lateral Prefrontal Cortex. Neuron 74(5):924–935

Pfùrtscheller G (1992) Event-related synchronization (ERS) - an Electrophysiological Correlate of Cortical Areas at Rest. Electroencephalogr Clin Neurophysiol 83(1):62–69

Prinzel LJ, Freeman FG, Scerbo MW, Mikulka PJ, Pope AT (2000) A closed-Loop System for examining psychophysiological measures for adaptive Task Allocation. Int J Aviat Psychol 10(4):393–410

Rheinberg F (2020) Intrinsic motivation and Flow. Motivation Sci 6(3):199–200

Sauseng P, Klimesch W, Doppelmayr M, Pecherstorfer T, Freunberger R, Hanslmayr S (2005) EEG alpha synchronization and functional coupling during top-down Processing in a Working Memory Task. Hum Brain Mapp 26(2):148–155

Shen X, Liu X, Hu X, Zhang D, Song S (2023) Contrastive learning of subject-invariant EEG representations for cross-subject emotion recognition. IEEE Trans Affect Comput 14(3):2496–2511

Thissen BAK, Schlotz W, Abel C, Scharinger M, Frieler K, Merrill J, Haider T, Menninghaus W (2022) At the heart of Optimal Reading experiences: Cardiovascular Activity and Flow experiences in Fiction Reading. Reading Res Q 57(3):831–845

Ulrich M, Keller J, Hoenig K, Waller C, Groen G (2014) Neural correlates of experimentally Induced Flow experiences. NeuroImage 86:194–202

Ursino M, Cesaretti N, Pirazzini G (2023) A model of Working Memory for Encoding multiple items and ordered sequences exploiting the Theta-Gamma Code. Cogn Neurodyn 17(2):489–521

Wang B, Han C, Wang T, Dai W, Li Y, Yang Y, Yang G, Zhong L, Zhang Y, Wu Y, Wang G, Yu H, Xing D (2021) Superimposed gratings induce diverse response patterns of Gamma oscillations in primary visual cortex. Sci Rep 11(1):4941

Wang Y, Han B, Li M, Li J, Li R (2023) An efficiently working Brain characterizes higher Mental Flow that elicits pleasure in Chinese calligraphic handwriting. Cereb Cortex 33(12):7395–7408

Weber R, Tamborini R, Westcott-Baker A, Kantor B (2009) Theorizing Flow and Media Enjoyment as Cognitive synchronization of attentional and reward networks. Communication Theory 19(4):397–422

Wollseiffen P, Schneider S, Martin LA, Kerherve HA, Klein T, Solomon C (2016) The Effect of 6 h of running on Brain Activity, Mood, and cognitive performance. Exp Brain Res 234(7):1829–1836

Woodruff CC, Daut R, Brower M, Bragg A (2011) Electroencephalographic α-band and β-band correlates of perspective-taking and personal distress. NeuroReport 22(15):744–748

Yazidi A, Mofrad AA, Goodwin M, Hammer HL, Arntzen E (2020) Balanced Difficulty Task Finder: an adaptive recommendation method for learning tasks based on the Concept of State of Flow. Cogn Neurodyn 14(5):675–687

Zhang Y, Wang F (2024) Developments and trends in Flow Research over 40 years: a bibliometric analysis. Collabra-Psychology 10(1):92948

Zhang W, Guo L, Liu D (2022) Transcerebral Information Coordination in Directional Hippocampus-Prefrontal Cortex Network during Working Memory based on bimodal neural electrical signals. Cogn Neurodyn 16(6):1409–1425

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.